DESCENT OF TOR-AMPLITUDE ALONG SURJECTIVE FLAT MORPHISMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Derived pullback preserves tor-amplitude. invoked at proof lines 3033-3034. Proves the only-if direction.
- Tor-amplitude in [a, b] can be checked on stalks. invoked at proof lines 3035-3036. Reduces to the local rings along f.
- A flat local ring homomorphism is faithfully flat. invoked at proof lines 3037-3038. Prepares the module-level descent input.
- Tor-amplitude of a complex descends along faithfully flat ring maps. invoked at proof lines 3039-3041. Concludes the if direction.

Source reference. Original lemma 09U8: sources/stacks-chow/tags.txt, source lines 12416–12416. Cited at proof lines 3033-3034 for the only-if direction; the descent argument is inside the counted span.

Source reference. Original lemma 09U9: sources/stacks-chow/tags.txt, source lines 12417–12417. Cited at proof lines 3035-3036 for the reduction to stalks; the descent conclusion is inside the counted span.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Cited at proof lines 3037-3038 to make the descent lemma applicable at stalks; the conclusion is inside the counted span.

Source reference. Original lemma 068S: sources/stacks-chow/tags.txt, source lines 8016–8016. Cited at proof lines 3040-3041 as the final module-level input; the conclusion is inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a surjective flat morphism of schemes (or more generally locally ringed spaces). Let $E \in D(\mathcal{O}_Y)$. Let $a, b \in \mathbf{Z}$. Then E has tor-amplitude in $[a, b]$ if and only if Lf^*E has tor-amplitude in $[a, b]$.*

Proof. Pullback always preserves tor-amplitude, see Cohomology, Lemma 09U8. We may check tor-amplitude in $[a, b]$ on stalks, see Cohomology, Lemma 09U9. A flat local ring homomorphism is faithfully flat by Algebra, Lemma 00HR. Thus the result follows from More on Algebra, Lemma 068S. □

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